

## Notulae Mycologicae (10)

auctore

Tsuguo HONGO

In this paper the total of eight species distributed in seven genera of the Agaricales are reported. Four of these, *Laccaria vinaceoavellanea*, *Amanita sychnopyramis* f. *subannulata*, *Tylopilus alutaceoumbrinus* and *Lactarius cyathula* f. *japonicus* have not previously been described; two are new to the Japanese flora; and two hitherto erroneously understood have been restudied.

Color terms given in quotation marks are from R. Ridgway, Color Standards and Color Nomenclature, Washington, D. C., 1912. Collections cited by number have been deposited in the writer's herbarium.

73) *Laccaria vinaceoavellanea* Hongo sp. nov. (Fig. 31, 1 & 2)

Pileo 4–6cm lato, convexo deinde expanso, interdum umbilicato vel depresso, non viscido, radiatim subsulcato-striato, furfuraceo, praesertim in medio, e brunneo-vinaceo avellaneo ("russet-vinaceous", "vinaceous-buff", etc.), sicco dilute alutaceo ("pale pinkish cinnamon", "pinkish buff", etc.); carne tenui, firma, concolore, sapore miti, odore leviter farinaceo; lamellis adnato-subdecurrentibus, distantibus ( $L=20-25$ ;  $l=(1)3-5$ ), 4–6mm latis, crassis, subconcoloribus; stipite 5–8cm longo, 6–8mm crasso, aequali, firmo, pileo concolore, fibroso-striato, basi albo-tomentoso, solido vel cavo; sporis in cumulo albis, sub microscopicis hyalinis, sphaeroideis,  $(5.5)7.5-8.5\mu$  diam., minute echinulatis, inamyloideis; basidiis tetrasporis; cystidiis nullis; tramate hymenophorali regulari.

Hab. Gregarious on the ground in forests, Mii-dera, Ôtsu-city, Aug. 27, 1952 (no. 456); Nangô-Imodani, Ôtsu, July 11, 1970 (no. 4160-type); Yabase, Kusatsu-city, Oct. 14, 1952 (no. 522); on sandy soil, Nakayama, Niigata-city, Aug. 28, 1955 (no. 1387, coll. I. Matsuda).

Distr. Japan.

Very common in both broad-leaved (*Castanopsis*, *Quercus*, etc.) and pine woods, in gardens, by roadsides, hedgerows, etc., from early summer to autumn. Resembling *L. laccata* (Fr.) Berk. & Br. most closely,

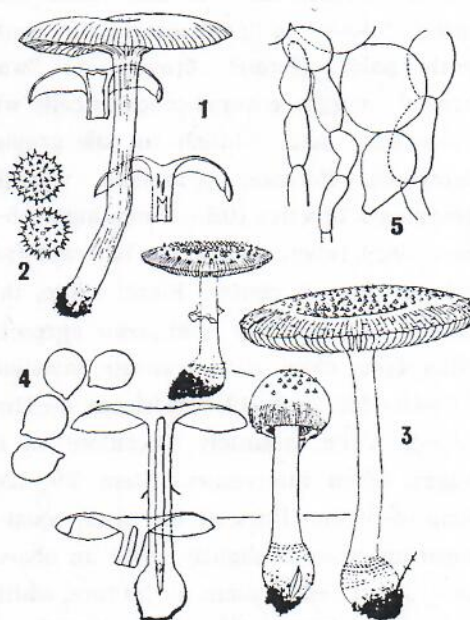


Fig. 31. *Laccaria vinaceoavellanea*: 1, carpophores; 2, spores. *Amanita sychnopyramis* f. *subannulata*: 3, carpophores; 4, spores; 5, crushed wart from cap. (1, 3  $\times 1/2$ ; 2, 4  $\times 750$ ; 5  $\times 450$ )

from which it differs in the dull color and in the somewhat sulcate-striate margin of the cap.

74) *Amanita sychnopyramis* Corner & Bas forma *subannulata* Hongo f. nov. (Fig. 31, 3—5)

Pileo 3–9cm lato, convexo, deinde expanso, medio subdepresso, in margine tuberculato-striato, obscure umbrino in centro, ad marginem pallidiore, verrucis conicis minutis albidis vel pallide griseo-brunneis oblecto; lamellis liberis, confertis, albis; stipite 3.5–12cm longo, 4–10mm crasso, deorsum obovate bulboso, albido, e farcto cavo; anulo candido, tenuissimo, subpersistente vel fugaci; volva adnata, subconcentrice squamuloso-diffracta; sporis globosis vel subglobosis,  $6.5-8\mu$  vel  $6.5-9 \times 6-7.5\mu$ , inamyloideis.

Cap 3–9cm broad, hemispherical to convex, then flattened with somewhat depressed center, surface more or less viscid when moist, "bister" to "clove brown" in center with paler ("snuff brown" to "wood brown") margin, covered concentrically with numerous small, whitish to pale grayish-brownish, floccose, pyramidal wart-like fragments of volva (0.5–1.5mm high, 0.5–2 mm wide), tuberculate-striate half-way from margin toward center. Flesh white, thin, rather fragile, taste mild, odor agreeable. Gills free, close, with truncate lamellulae ( $L=80-102$ ;  $I=0-1$ ), ventricose, 4–11mm broad, white, minutely flocculose on the edges, often intervenose. Stem 3.5–12cm long, 4–10mm thick at the apex, equal or tapering upward slightly, with an obovate bulb about 1–2cm diam. at the base, whitish to pale pinkish buff, often grayish near the base, the upper part of the bulb set with ivory yellow, small, wart-like volval fragments, arranged in more or less concentric encircling rings, stuffed or

hollow. Ring white, small, very thin, with grayish flocculose margin, somewhat persistent or fugacious (and at times observed only in unexpanded specimens). Spores hyaline under the microscope, globose to subglobose,  $6.5-8\mu$  or  $6.5-9 \times 6-7.5\mu$ , thin-walled, smooth, nonamyloid; basidia four-spored,  $30-44 \times 11-13\mu$ ; marginal cells clavate, piriform or subglobose,  $20-50 \times 10-26\mu$ ; fragments of volva on cap consisting of more or less erect chains of ellipsoid, globose, clavate or subcylindric, thin-walled cells,  $17-38 \times 10-31\mu$ , mixed with  $5-7.5\mu$  broad hyphae.

Hab. Gregarious on the ground in forest of *Castanopsis cuspidata*, Kokubu, Ôtsu-city, July 3, 1970 (no. 4139); July 15, 1970 no. 4166-type).

Distr. Japan (Shiga).

In evergreen broad-leaved woods, probably mycorrhizal with *Castanopsis*; not very common, in summer.

*A. sychnopyramis* was first described from Singapore by Corner & Bas (Persoonia 2: 291. 1962). The writer's plant differs from the original description in the presence of the partial veil, but otherwise the diagnoses square very well. So the writer does not hesitate to regard it as a subannulate form of *A. sychnopyramis*.

Somewhat resembling *A. pantherina* (Fr.) Secr., but differing chiefly in the smaller size, the shape of warts on the cap and the smaller, spherical spores.

75) *Agaricus abruptibulbus* Peck, N. Y. State Mus. Bull. 94: 36. 1905. (Fig. 32, 1–3)

Cap 5–11cm broad, convex then expanded-plane, surface dry, silky-fibrillose, sometimes more or less appressed-scaly, pale yellow ("massicot yellow"), with whitish margin, often with dingy yellowish stains, margin at first incurved. Flesh moderately

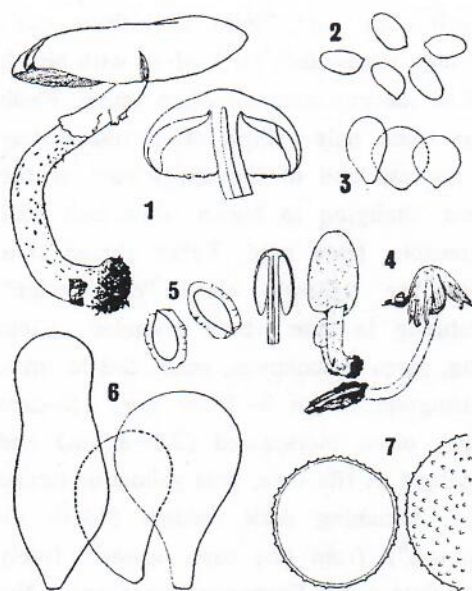


Fig. 32. *Agaricus abruptibulbus*: 1, carphores; 2, spores; 3, marginal cells. *Coprinus narcoticus*: 4, carphores; 5, spores; 6, cheilocystidia; 7, globular cells on the cap. (1, 4  $\times 1/2$ ; 2, 5  $\times 750$ ; 3, 6  $\times 450$ ; 7  $\times 300$ )

thick, whitish, unchangeable in the cap, in the stem becoming pale ochraceous-buff with age; odor slight, agreeable. Gills free, and remote from the stem, at first whitish, then avellaneous, finally purplish brown, rather narrow (4–8mm), crowded. Stem 9–13cm long, 1–1.5cm thick, equal or slightly tapering upward from an abrupt bulb, covered by scattered white floccose squamules or fibrils below the ring, white, becoming pale pinkish cinnamon with age, hollow. Ring broad, thin,  $\pm$ double, white or pale yellowish above, adorned below with floccose patches, superior. Spores grayish brown under the microscope, elliptic, smooth  $6.5\text{--}7.5 \times 3.5\text{--}5\mu$ ; marginal cells oval, subspherical or pear-shaped, thin-walled,  $14\text{--}23 \times 8\text{--}20\mu$ .

Hab. Gregarious or solitary, among fallen leaves in bamboo forests, Shinhami, Kusatsu-city, Sept. 20, 1945 (no. 7); Sept. 23, 1952 (no. 488): Kamidanakami-Shibahara,

Ôtsu-city, July 1, 1970 (no. 4138): under *Quercus acutissima*, Nangô-Imodani, Ôtsu, Aug. 10, 1968 (no. 3692).

Distr. North America, Europe, Japan (Shiga, Kyoto).

Ill. Atkinson, Mushrooms, figs. 19, 20 (sub nom. *silvicola*); Kauffman, Agar. Mich. pl. 47; J. Lange. Fl. Agar. Dan. pl. 138, B (sub nom. *silvicola*); Wakefield & Dennis, Comm. Brit. F. pl. 73, fig. 1 (sub nom. *silvicola*); Imazeki & Hongo, Col. Ill. F. Jap. 1: pl. 22, fig. 117 (sub nom. *arvensis*); Essette, Les Psalliotes, pl. 33.

In deciduous, mixed or bamboo forests, from summer to autumn, not uncommon. This fungus has hitherto been confused with *A. silvicola* (Vitt.) Peck, which differs in lacking a distinct bulb and in its smaller spores ( $5\text{--}6 \times 3\text{--}4\mu$ ).

76) *Coprinus narcoticus* (Fr.) Fr. Epicr. 250. 1838. (Fig. 32, 4–7)

Cap at first oval, 1.5–2cm high, 1–1.3cm broad, then campanulate to flattened, finally with the edge upturned and split (1.5–2.5cm broad), entirely covered by a dense coating of whitish to pale grayish (“pale drab-gray”) meal which is made up of globular cells. Flesh very thin at the margin, watery grayish, odor very strong and foetid. Gills free, white, then dark reddish brown and finally blackish, about 3mm broad. Stem 2–6cm long, 1–3.5mm thick, whitish to pale grayish, mealy to villous-downy, fragile, fistulose. Spores blackish brown in mass, gray-brown under the microscope, elliptic,  $8\text{--}11 \times 4.5\text{--}5.5\mu$ , with a hyaline epispore ( $10\text{--}13.5 \times 6.5\text{--}8\mu$  including epispore); basidia four-spored; cheilocystidia abundant,  $50\text{--}87 \times 30\text{--}45\mu$ , spheropedunculate or vesiculose-fusiform, thin-walled, hyaline; cells of mealy coating spherical,  $33\text{--}97\mu$  in diameter, minutely granulate.

Hab. Gregarious on decaying straw-mat, Tanakami-Ishizue, Ôtsu-city, May 8, 1971 (no. 4373).

Distr. Europe, North Africa. New to Japan (Shiga).

Ill. Cooke, Ill. Brit. F. pl. 668 (680) B; Konr. & Maubl. Ic. Sel. F. pl. 36 II; J. Lange, Fl. Agar. Dan. pl. 159 D; Kühn. & Romagn. Fl. Anal. Champ. Supér. 385, fig. 536(spore).

Easily recognizable by the whitish to grayish, mealy cap, the strong disagreeable smell and the spores having a hyaline covering. Probably not uncommon.

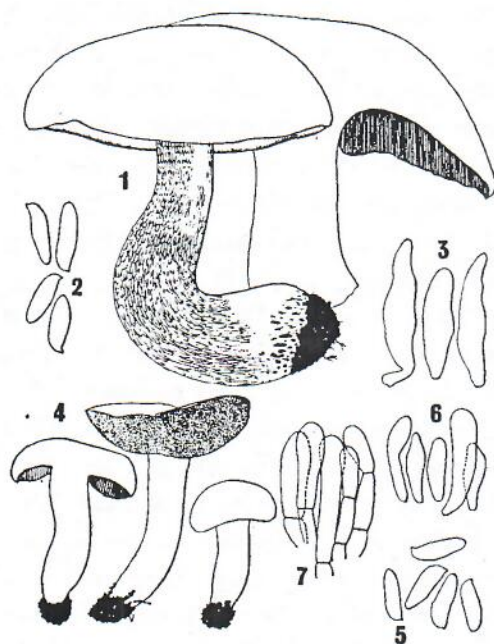


Fig. 33. *Boletus speciosus*: 1, carpophores; 2, spores; 3, cystidia. *Tylopilus alutaceoum-brinus*: 4, carpophores; 5, spores; 6, cystidia; 7, a part of cuticle. (1, 4  $\times 1/2$ ; 2, 5  $\times 750$ ; 3, 6, 7  $\times 450$ )

77) *Boletus speciosus* Frost, Bull. Buff. Soc. Nat. Science 2: 101. 1874—*Boletus regius* Krombh. s. Kawam., Imaz., Imaz. & Hongo non al. (Fig. 33, 1—3)

Cap 8—11cm broad, convex, pulvinate, surface almost smooth, subviscid in wet weather, rose pink (somewhat brighter than

“Corinthian red”, “light Corinthian red”, “orange vinaceous”, etc.), often with bluish stains, margin incurved when young. Flesh very thick, pale yellow, often tinged rosy in the cap and in the upper part of the stem, changing to bluish when cut, odor agreeable, taste mild. Tubes almost free, “primrose yellow”, then “clay color”, changing to blue where wounded,  $\pm 1$ cm long; pores concolorous, small, 2–3 to 1mm, subangular. Stem 9–10cm long, 1.5–2cm thick, often incrassated (2.5–3.5cm) and incurved at the base, pale yellow or tinged rosy, becoming dark reddish (“dark vinaceous”) from the base upward, finely reticulate, solid. Spores melleous under the microscope,  $10\text{--}13.5 \times 3\text{--}3.5\mu$ , subfusoid-subcylindric, smooth; cystidia  $27\text{--}51 \times 9.5\text{--}11.5\mu$ , fusoid-subcylindric or almost flask-shaped, melleous.

Hab. In *Pinus densiflora*-*Quercus serrata* forest, Wachi-chô, Kyoto-pref., June 29, 1970 (no. 4136, coll. T. Ito); in mixed forest (*Abies firma*, *Fagus japonica*, *Quercus mongolica* var. *grosseserrata*, etc.), Misugimura, Mie-pref., Aug. 26, 1971 (no. 4524).

Distr. North America, Europe, Japan.

Ill. Farlow, Icon. Farlow. pl. 81; Singer, Die Röhrlinge, 2: pl. 8, figs. 1–7; Coker & Beers, Boletac. N. Carolina, pls. 44 (above) & 64, fig. 5; Snell & Dick, The Boleti, pls. 39, 39a & 83, fig. 1; Kawamura, Icon. Jap. F. 2: fig. 240 (sub nom. *regius*); Imazaki, “Genshoku-Kinoko”, 177 (sub nom. *regius*); Imazeki & Hongo, Col. Ill. F. Jap. 2: pl. 30, fig. 188 (sub nom. *regius*).

In Japan, this species was no doubt confused until recent times with *B. regius* Krombh., which has the purer and brighter pink or purplish pink cap and the broader spores ( $11\text{--}15 \times 4\text{--}5\mu$ ). From summer to autumn, often under beeches, oaks, etc.,

not very common.

78) *Tylopilus alutaceoumbrinus* Hongo  
sp. nov. (Fig. 33, 4–7)

Pileo 3–6.5cm lato, convexo, pulvinato, dein expanso, sicco, minute tomentoso-velutino vel subglabro, alutaceo-umbrino ("sayal brown", "snuff brown", "tawny-olive", "Saccardo's umber" vel "sepia"), margine primum incurvata; carne crassa, subfirma, alba, fracta subrubescens, odore nullo, sapore amaro; tubulis adnatis vel depressis, albidis dein pallide carneis, 4–7mm longis; poris subangulatis, parvis, 2–3 in uno mm, concoloribus, tactu sordide brunnescentibus; stipite 3–6cm longo, 6–12mm crasso, subaequali vel subventricoso, sicco, pileo concolore, minute velutino, solido; sporis in cumulo pallide vinaceo-incarnatis (e "pale vinaceous-pink" "light vinaceous-fawn"), sub microscopio dilute melleis, subfusoidis vel subcylindricis, laevibus,  $8.5-11 \times 3-3.5 \mu$ ; basidiis tetrasporis; cystidiis numerosis, praecipue ad poros,  $20-34 \times 5-8.5 \mu$ , hyalinis vel aureis, subfusoidis, clavatis vel subcylindricis; tramate hymenophorali laterali typi *Boletorum*; epicute trichodermium palisadicum formante.

Hab. Gregarious or solitary, on the ground in forest of *Castanopsis cuspidata*, Mii-dera, Ôtsu-city, Sept. 8, 1962 (no. 2538): in mixed forest of *Pinus densiflora*, *Quercus serrata*, etc., Mt. Tanakami-yama, Ôtsu, Aug. 8, 1968 (no. 3687): in *Pinus-Quercus* forest, Kamidanakami-Shibahara, Ôtsu, Aug. 12, 1968 (no. 3694-type); Aug. 20, 1969 (no. 3970); July 20, 1970 (no. 4176).

Distr. Japan (Shiga).

Not uncommon from summer to autumn. Differs from *T. otsuensis* Hongo in the more brownish color, bitter taste and elongated spores. *T. indecisus* (Peck) Murr. is somewhat similar in color, but differs in

the reticulate stem, mild taste and larger spores.

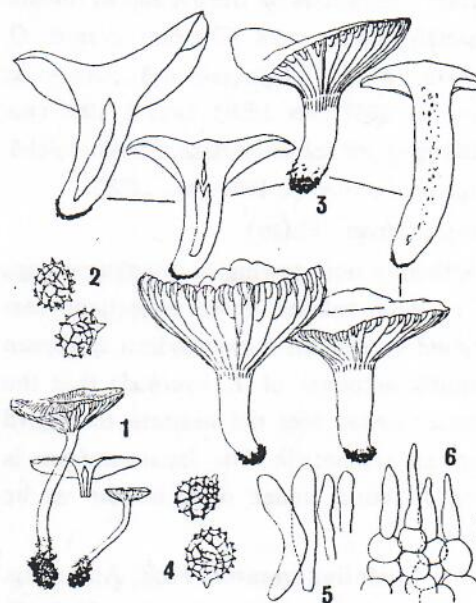


Fig. 34. *Lactarius cyathula* f. *japonicus*: 1, carpophores; 2, spores. *Lactarius gerardii*: 3, carpophores; 4, spores; 5, cheilocystidia; 6, cap cutis. (1, 3  $\times 1/2$ ; 2, 4  $\times 750$ ; 5, 6  $\times 450$ )

79) *Lactarius cyathula* (Fr.) Fr. forma *japonicus* Hongo f. nov. (Fig. 34, 1 & 2)

A typo differt habitatione non sphagnicola.—In silvis frondosis mixtisque.

Cap 1–3cm broad, flattened then depressed, usually papillate, surface glabrous, sordid orange-brown, darker (umbrinous) in the center, radially rugulose, somewhat translucent-striate when moist. Flesh very thin, concolorous with the surface, odor none. Milk white, scanty, taste mild. Gills adnate or adnato-subdecurrent, distant, unbranched, 1–4mm broad, creameous, then ochraceous-buff to flesh color. Stem 1–5cm long, 1–3mm thick, slender, often flexuose, equal, glabrous, reddish brown, paler at the apex, hollow, the base often covered with whitish or tawny hairs. Spores pale cream in mass, subspheric oval,  $8.5-10 \times 7-8 \mu$ ,

warty-spinulose and subreticulate; basidia four-spored,  $33-40 \times 10-12\mu$ .

Hab. Gregarious on the ground in forests, especially under oaks (*Quercus glauca*, *Q. serrata*, etc.), Ishiyama-Senjô, Ôtsu-city, Sept. 20, 1957 (no. 1658); Oct. 8, 1958 (no. 1918-type); Mt. Shizuga-dake, Kinomoto-chô, Shiga-pref., Nov. 3, 1958 (no. 1958).

Distr Japan (Shiga).

Although not growing in *Sphagnum* bogs or in damp places, these collections correspond so well with the modern European author's accounts of *L. cyathula* that the present writer does not hesitate to regard them as conspecific. The Japanese form is usually found under oaks as far as he knows.

80) *Lactarius gerardii* Peck, Ann. Rep. N. Y. State Mus. 26: 65. 1874. (Fig. 34, 3-6)

Cap 3-7.5cm or more broad, convex, then plane and depressed, with or without a central conic umbo, surface dry, radially rugulose, azonate, pruinously velvety under a lens, "snuff brown", darker ("bister") at the center, margin sometimes plicate. Flesh firm, thin, white, not changing color, odor not distinctive, taste mild. Milk white, unchanging. Gills at first broadly adnate becoming decurrent in age, distant, intervenose, 5-10mm broad, white or pale cream color, becoming slowly brownish where wounded, the edge often dark brown, especially near the margin of the cap. Stem 3-6cm long, 8-13mm thick, equal or somewhat tapering downward, often plicate at the apex, concolorous with the cap, pruinously velvety under a lens, white tomentose at the base, stuffed becoming hollow. Spores white in mass,  $7.5-9 \times 6.5-7.5\mu$  or  $8-10.5 \times 7-9\mu$ , subspheric, reticulate with spines; basidia four-spored,  $39-48 \times 10-11\mu$ ; cheilocystidia sparse or abundant, filamentous,

subfusoid or subcylindric,  $28-41 \times 3.5-9\mu$ ; epicutis of cap 3 or more cells deep, of enlarged cells, the uppermost bearing cystidia-like outgrowths which are filamentous or narrowly fusoid and  $23-33 \times 2.5-6\mu$ , content of all cells umbrinous in KOH.

Hab. Gregarious to scattered on the ground in forests of *Castanopsis cuspidata*, Mii-dera, Ôtsu-city, Aug. 6, 1951 (no. 203); Aug. 21, 1952 (no. 453); Sept. 1, 1952 (no. 460), Aug. 26, 1953 (no. 720); July 17, 1954 (no. 949); Ishiyama-dera, Ôtsu, Aug. 13, 1952 (no. 445); Arato-jinja, Ôtsu, Aug. 21, 1969 (no. 3974); Tanakami-Sekinotsu, Ôtsu, Sept. 14, 1969 (no. 4001); in *Pinus densiflora*-*Quercus serrata* forest, Tanakami-Ishizue, Ôtsu, July 21, 1961 (no. 2303); July 16, 1965 (no. 3102).

Distr. Eastern regions of North America. New to Japan (Shiga).

Ill. Smith & Hesler, Brittonia 14: pl. 16 & p. 433, fig. 73. 1962.

Rather common, from summer to autumn. Distinguished from *L. lignyotus* Fr. by the very distant gills, white spore deposit and absence of any reddish staining of the flesh when cut.

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## 菌 類 記 (10)

この報文には日本産ハラタケ目菌類8種を載録した。引用した標本はすべて滋賀大学教育学部生物学区の筆者の手もとに保存されている。

73) *Laccaria vinaceoavellanea* Hongo カレバキツネタケ (新種)。ごく普通にみられるキツネタケ近似種で、かさや茎がくすんだ色あいをおび、かさの表面には放射状のみぞ線があらわれる。筆者所蔵の標本は天津市、草津市、新潟市などで採集されたもので、おそらく全国に広く分布するものと思われる。

74) *Amanita sychnopyramis* Corner & Bas f. *subannulata* Hongo テングタケダマシ (新品種)。シンガポールから報告された *A. sychnopyramis* の1品種と考えられるもので、膜質のうすい内被膜を有する点以外は全く一致している。テングタケ *A. pantherina* (Fr.) Secr. を小形にしたような観があるが、外被膜の破片が錐状であること、胞子が小さく球形であること、などで区別される。産地：大津市国分シイ林内。

75) *Agaricus abruptibulbus* Peck ウスキモリノカサ (新称)。従来欧米でしばしば *A. silvicola* (Vitt.) Peck と同定されていた種類である。また、今関・本郷：原色日本菌類図鑑, pl. 22, fig. 117 に図説された「シロオオハラタケ」も本種である。夏から秋にかけて、クスギ・コナラ林、竹林などに発生する。筆者所蔵の標本は草津市および大津市で採集されたものである。

76) *Coprinus narcoticus* (Fr.) Fr. ニオイコナヒトヨタケ (新称)。小形のヒトヨタケで表面は粉 (球形細胞) におおわれ、強い臭気がある。胞子が明りょうな外膜を有することは特徴的である。産地：大津市田上石居 (たなかみいしずえ) 町、アカマツ林内に捨てられた古だたみ上。

77) *Boletus speciosus* Frost アカジコウ (川村), アケボノヤマドリタケ (今関)。従来日本で *B. regius* Kormbh. と同定されていた

キノコであるが、上記の学名を当てるのが妥当と考える。*B. regius* のかさの色は明るいピンク色または帯紫淡紅色で、胞子はやや幅が広い。産地：京都府和和町、アカマツ・コナラ林 (京都府林業試験場、伊藤武技師採集) および三重県美杉村三重大学平倉演習林、モミ・広葉樹の混生林。

78) *Tylopilus alutaceoumbrinus* Hongo フモトニガイグチ (新種)。ややコビチャニガイグチ *T. otsuensis* Hongo に似ているが、多少色あいが異なり、苦味があり、また胞子が長形である。夏から秋にかけて、シイ林やアカマツ・コナラ林などに発生する。大津市内諸所で採集。

79) *Lactarius cyathula* (Fr.) Fr. f. *japonicus* Hongo アキノアシボソチチタケ (新品種)。形態的には *L. cyathula* とよく一致するが、生態的に異なるのでその1品種とみなした。*L. cyathula* は湿原のハンノキ類の樹下に発生するといわれている。産地：大津市石山千町および滋賀県木之本町賤が岳の林内、アラカシ・コナラなどの樹下。

80) *Lactarius gerardii* Peck クロチチタケ (新称)。クロチチタケ *L. lignyotus* Fr. の一群 (sect. *Plinthogali*) に属する種類であるが、肉または乳液は空気にふれても紅変することはない。大津市内諸所のシイ林、アカマツ・コナラ林などで採集。おそらく西日本に広く分布するものと思われる。

